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Kelas : 1A-TNK

No.
Date:

1. Ubahlah ukuran sudut berikut ke dalam bentuk desimal!

a) $21°45'30''$ b) $45°15'$ c) $61°42'21''$

d) $37°11'29''$ e) $128°35'18''$

jawab

a) $21° 45' 30'' = 21° + \frac{45}{60} + \frac{30}{3600} = 21° + 0,75° + 0,008333°$

$= 21,75833°$

b) $45°15' = 45° + \frac{15}{60} = 45° + \frac{1}{4} = 45,25°$

c) $61°42'21'' = 61 + \frac{42}{60} + \frac{21}{3600} = 61° + 0,7° + 0,005833°$

$= 61,70583°$

d) $37°11'29'' = 37° + \frac{11}{60} + \frac{29}{3600} = 37° + 0,183333° + 0,008055°$

$= 37°, 19138°$

e) $128°35'18° = 128 + \frac{35}{60} + \frac{18}{3600} = 128° + 0,58333° + 0,005°$

$= 128,58833°$

2 Ubahlah ukuran sudut ke D°M'S" !

a) $61.24°$ b) $28.6582°$ c) $165.8525°$

d) $225.24°$ e) $-76.24°$

Jawab

a) $61,24°$

- $D = 61°$
- $M = 0,24 \times 60 = 14,4° \rightarrow 14°$ hasilnya $61°14'24''$
- $S = 0,4 \times 60 = 24''$

b) $28,6582$

- $D = 28°$ hasilnya $28°39'30''$
- $M = 0,6582 \times 60 = 39,492'$
- $S = 0,492 \times 60 = 29,52'' \approx 30''$

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No.

Date:

c. $165,8525^\circ \rightarrow D = 165^\circ$

$M = 0,8525 \times 60 = 51,15'$

$S = 0,15 \times 60 = 9''$

hasilnya $165^\circ 51' 9''$

d. $225,24^\circ \rightarrow D = 225^\circ$

$M = 0,24 \times 60 = 14,4'$

$S = 0,4 \times 6 = 24''$

hasilnya $= 225^\circ 14' 24''$

e. $-76,24^\circ \rightarrow D = 76^\circ$

$M = 0,24 \times 60 = 14,4'$

$S = 0,4 \times 60 = 24''$

hasilnya $-76^\circ 14' 24''$

3. Ubahlah ukuran sudut berikut ke radian!

a) $30^\circ = \frac{30^\circ \pi}{180} = \frac{1}{6}\pi$ rad

b) $60^\circ = \frac{60^\circ \pi}{180^\circ} = \frac{1}{3}\pi$ rad

c) $135^\circ = \frac{135^\circ \pi}{180^\circ} = \frac{3}{4}\pi$ rad

d) $240^\circ = \frac{240^\circ}{180^\circ}\pi = \frac{4}{3}\pi$ rad

e) $-60^\circ = -\frac{\pi}{3}$ rad

4. ubahlah ukuran sudut ke dalam radian!

a) $25^\circ = \frac{25^\circ \pi}{180^\circ} = \frac{5}{36}\pi$ rad $= 0,436$ rad

b) $36^\circ = \frac{36^\circ}{180^\circ} \cdot \pi = \frac{1}{5}\pi$ rad $= 0.628$ rad

c) $212^\circ = \frac{212^\circ}{180^\circ}\pi = \frac{53}{45}\pi$ rad $= 3.700$ rad

d) $25^\circ 15' 35'' = 25 + \frac{15}{60} + \frac{35}{3600}$

$= 25 + 0.25 + 0,009722 = 25.2597^\circ$

konversi $25.2597 \cdot \frac{\pi}{180} = 0,14033166\pi$ rad $= 0.441$ rad

e) $240^\circ 45' = 240^\circ + \frac{45}{60} = 240.75^\circ$

konversi $240.75^\circ \times \frac{\pi}{180} = 1,3375\pi$ rad $= 4.201$ rad

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No.

Date:

5. Ubahlah ukuran sudut berikut ke derajat (degress)!

a) $0,42 \text{ rad} \times \frac{180}{\pi} = 24,06^\circ \approx 24,1^\circ$

b) $1,3 \text{ rad} \times \frac{180}{\pi} = 74,48^\circ \approx 74,5^\circ$

c) $3,5 \text{ rad} \times \frac{180}{\pi} = 200,53^\circ \approx 200,5^\circ$

d) $\frac{3\pi}{5} \cdot \frac{180}{\pi} = 108^\circ$

e) $\frac{7\pi}{9} \times \frac{180}{\pi} = 140^\circ$

6. Solve trigonometri dengan x antara 0° dan 360°

1) $8 \sin x = 4$

$\sin x = \frac{1}{2}$

$x = 30^\circ$ atau $x = 150^\circ$

2) $6 - 12 \cos x = 0$

$-12 \cos x = -6$

$\cos x = \frac{1}{2}$

$x = 60^\circ$ atau $x = 300^\circ$

3) $20 \sin x + 10\sqrt{3} = 0$

$20 \sin x = -10\sqrt{3}$

$\sin x = \frac{-10}{20}\sqrt{3}$

$\sin x = -\frac{1}{2}\sqrt{3}$

$x = 240^\circ$ atau $x = 300^\circ$

4) $8\sqrt{3} \tan x + 8 = 0$

$8\sqrt{3} \tan x = -8$

$\tan x = \frac{-8}{8\sqrt{3}} = -\frac{1}{3}\sqrt{3}$

$x = 150^\circ$ atau $x = 330^\circ$

5) $10 \sin (x + 30^\circ) = 5$

$\sin (x + 30^\circ) = \frac{1}{2}$

$x + 30^\circ = 30^\circ$ atau $x + 30 = 150^\circ$

$x = 0^\circ$ $\quad x = 120^\circ$

6) $6 \cos (x + 60^\circ) = 5$

$\cos (x + 60^\circ) = 5/6$ $\quad 5/6 = 0,834$

$\arccos 0,834 = 33,56$

$x + 60^\circ = 33,56^\circ$

$x + 60^\circ = 360^\circ - 33,56^\circ = 326,44^\circ$

$x = 266,44^\circ$ atau $x = 33,56^\circ$

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Date:

7. $12\cos(x-\pi/4)=6$

$\cos(x-\pi/4)=\frac{1}{2}$

$\cos(x-\pi/4)=\cos\pi/3$

$x-\pi/4=\pi/3$ atau $x-\pi/4=5/3\pi$

$x=\frac{7}{12}\pi$ atau $x=\frac{23}{12}\pi$

$x=105^0$ $x=345^0$

8. $20\sin(x-0,132)=10$

$\sin(x-0,132)=1/2$

$\sin(x-0,132)=\sin 30^0$

$x-0,132=\pi/6$ atau $x-0,132=5/6\pi$

$x=\pi/6+8/25$ $x=5\pi/6+8/25$

9. $4\sin^2x-3=0$

$\sin^2x=3/4$

$\sin x=\pm\sqrt{3/4}$

$\sin x=\frac{\sqrt{3}}{2}$ atau $\sin x=-\frac{\sqrt{3}}{2}$

$x=\pi/3$ rad $x=\frac{4\pi}{3}$ rad atau $\frac{5\pi}{3}$ rad

atau $x=\frac{2}{3}\pi$ rad